

1. Executive Summary

		Traffic Summary		Count	
		Total Flows Observed		14994	
Total Threats Mitigated		18		6184211209	
High Priority		1		6169564146	
Low Priority		17		10136	
Manual Releases		0			
Manual Blocks		0			
ML Metrics		Value		Attack Vector	
False Positives		1		ICMP Flood	
FP Rate		0.01%		0	
IF Precision		99.98%		SYN Flood	
IF Recall		90.62%		14	
IF F1-Score		95.07%		UDP Flood	
IF Accuracy		96.64%		0	
				Uncertain	
				4	
				Action	
				Quarantined	
				4	
				Time Ban	
				4	
				Blackhole	
				0	
				Released	
				2	

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	99.98%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	90.62%	Share of actual attacks successfully flagged
F1-Score	95.07%	Balanced measure combining Precision and Recall
Accuracy	96.64%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.01%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	9.38%	Actual attacks that went undetected
True Positive Rate (TPR)	90.62%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.99%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 4857	FN 503
Actual: Normal	FP 1	TN 9633

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	100.00%	Share of actual attacks successfully flagged
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	0.00%	Actual attacks that went undetected
True Positive Rate (TPR)	100.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	100.00%	Normal traffic correctly identified as safe

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	4857	0	0
Act: ICMP	0	0	0
Act: UDP	0	0	0

2c. Hybrid Cascaded

Metric	Value	Description
Precision	99.98%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	90.62%	Share of actual attacks successfully flagged
F1-Score	95.07%	Balanced measure combining Precision and Recall
Accuracy	96.64%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.01%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	9.38%	Actual attacks that went undetected
True Positive Rate (TPR)	90.62%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.99%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 4857	FN 503
Actual: Normal	FP 1	TN 9633

2d. Response Latency

Metric	Value	Description
Detection Time	633.7 ms	Average time taken to detect an attack
Mitigation Response Time	13.7 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2e. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	1.00%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	4.35%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	13.02%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.16	3	3	2026-09-16 12:07:57	2026-09-16 12:09:20
10.0.0.19	3	3	2026-09-16 12:08:08	2026-09-16 12:09:30
10.0.0.17	2	2	2026-09-16 12:08:04	2026-09-16 12:09:15
10.0.0.18	2	2	2026-09-16 12:07:58	2026-09-16 12:09:09

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-16 12:07:57	10.0.0.16	DDoS	Uncertain	-	Low	Quarantined
2026-09-16 12:07:58	10.0.0.18	DDoS	Uncertain	-	Low	Quarantined
2026-09-16 12:08:05	10.0.0.17	DDoS	Uncertain	-	Low	Quarantined
2026-09-16 12:08:08	10.0.0.16	DDoS	SYN Flood	93.0%	Critical	Time Ban (1m)
2026-09-16 12:08:08	10.0.0.19	DDoS	Uncertain	-	Low	Quarantined
2026-09-16 12:08:09	10.0.0.18	DDoS	SYN Flood	90.5%	Critical	Time Ban (1m)
2026-09-16 12:08:10	10.0.0.18	DDoS	SYN Flood	90.5%	Critical	Time Ban
2026-09-16 12:08:11	10.0.0.16	DDoS	SYN Flood	93.0%	Critical	Time Ban
2026-09-16 12:08:15	10.0.0.17	DDoS	SYN Flood	92.5%	Critical	Time Ban (1m)
2026-09-16 12:08:16	10.0.0.17	DDoS	SYN Flood	92.5%	Critical	Time Ban
2026-09-16 12:08:18	10.0.0.19	DDoS	SYN Flood	93.1%	Critical	Time Ban (1m)
2026-09-16 12:08:18	10.0.0.19	DDoS	SYN Flood	93.1%	Critical	Time Ban
2026-09-16 12:09:08	10.0.0.16	DDoS	SYN Flood	93.7%	High	Released
2026-09-16 12:09:09	10.0.0.18	DDoS	SYN Flood	92.6%	Critical	Time Ban (2m)
2026-09-16 12:09:15	10.0.0.17	DDoS	SYN Flood	92.5%	Critical	Time Ban (2m)
2026-09-16 12:09:18	10.0.0.19	DDoS	SYN Flood	93.1%	Critical	Released
2026-09-16 12:09:20	10.0.0.16	DDoS	SYN Flood	93.7%	Critical	Time Ban (2m)
2026-09-16 12:09:30	10.0.0.19	DDoS	SYN Flood	93.1%	Critical	Time Ban (2m)

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.19	SYN Flood	0.6513	93.1%	Critical	Phase 2	3	11s	2026-09-16 12:09:30	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6349	93.7%	Critical	Phase 2	3	11s	2026-09-16 12:09:20	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6409	93.1%	Critical	Phase 2	1	1m 10s	2026-09-16 12:09:18	Ban Expired
10.0.0.17	SYN Flood	0.6234	92.5%	Critical	Phase 2	2	1m 11s	2026-09-16 12:09:15	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6376	92.6%	Critical	Phase 2	2	1m 11s	2026-09-16 12:09:09	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.5965	93.7%	High	Phase 2	1	1m 11s	2026-09-16 12:09:08	Ban Expired
10.0.0.19	SYN Flood	0.6511	93.1%	Critical	Phase 2	1	10s	2026-09-16 12:08:18	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.17	SYN Flood	0.6367	92.5%	Critical	Phase 2	1	11s	2026-09-16 12:08:15	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6372	90.5%	Critical	Phase 2	1	11s	2026-09-16 12:08:09	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6383	93.0%	Critical	Phase 2	1	11s	2026-09-16 12:08:08	Attack persisted - escalating to time ban

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME
Network Administrator

DATE